



Dealer Scheduling — Web Automation Proof of Concept

Pilot dealer: Fox Taylor. Goal: book a real service appointment end-to-end through the dealer's existing public website.

PILOT DEALER	APPROACH	EFFORT ESTIMATE	OUTPUT
Fox Taylor	Web scraping / browser automation	~2 weeks	POC report + repeatability demo

1. Project

Kate (the agent) needs to actually *book* the appointment she promises the customer. Until dealer systems expose us a real API, the only way to do that reliably is to drive the dealer's public website the way a human would. This work package proves we can do exactly that against a single live dealer — Fox Taylor — and lays the pattern for adding more dealers later.

Objectives

For the Fox Taylor service scheduler, programmatically demonstrate the five steps of a real booking flow:

STEP 1 Access the URL

Reach the Fox Taylor service scheduling page from a clean session, no human in the loop.
Identifies: hosting platform (Xtime, MyKaarma, Tekion, dealer's own CMS), cookie/consent walls, geo restrictions, rate limits.

STEP 2 Navigate through it

Walk the booking funnel: vehicle identification (VIN or year/make/model + license/mileage), customer info entry, service selection page, calendar page, confirmation page.
Captures the page graph and form structure so the bot can resume mid-flow if interrupted.

STEP 3 Fetch open slots

Read the dealer's available appointment times for a target date range and return them as a structured list (date, time, advisor if shown, capacity if exposed).
This is the input Kate uses when she asks the customer "Tuesday at 8:30 or Wednesday at 1?" — it must reflect real dealer inventory, not stale or fabricated data.

STEP 4 Select the service

Map our internal service codes (recall ID, DTC code, standard maintenance interval) onto the dealer's own service catalog as displayed on the page.
Dealers describe services differently ("60K major service" vs "60,000 mile interval" vs "Service B"). The mapping table is part of the deliverable.

STEP 5 Book the service

Submit the booking. Capture the dealer's confirmation number, time, advisor, and any email/SMS the dealer sends to the customer. Verify the appointment shows up on the dealer side.
This is the "moment of truth" — the only step that materially changes dealer state. We need it logged, idempotent, and reversible (cancel flow as a follow-up).

Out of scope (this work package)

- Multi-dealer abstraction — Fox Taylor only.
- Cancel / reschedule flows — those become Work Package 02 once Book is proven.
- Production hardening (proxies, captcha service, retry queues) — POC runs from one machine.
- Direct API integration — only revisited if the dealer offers one.

2. Approach

Tooling

- Playwright** (Python) for headless-Chromium driving — handles JavaScript-heavy SPAs, modern dealer schedulers (Xtime, etc.) won't work with plain HTTP requests.
- Page object model** — one Python class per page (`VehicleEntryPage` , `ServicePickerPage` , `CalendarPage` , `ConfirmPage`). Selectors centralized so a dealer site change is one file to fix.
- Recording + replay** — Playwright's trace viewer captures every action with screenshots; this is the audit trail when something breaks.
- Test data** — a small set of real customer + vehicle records supplied by Fox Taylor (with their consent) so we book real appointments we will then cancel by phone.

Phases

- Reconnaissance (1–2 days)** — manually walk the Fox Taylor scheduler, identify the underlying platform, capture network traffic, document every form, every required field, every async call.
- Read-only build (3–4 days)** — automate Steps 1–3. At this point we can fetch real open slots without changing any dealer state. Safe to run repeatedly.
- Write build (3–4 days)** — automate Steps 4–5. First booking is reviewed live by an operator before submit. Subsequent bookings progressively unattended.
- Repeatability run (1 day)** — five back-to-back bookings with different vehicles, different services, different time windows. All canceled by phone after.

3. Deliverable A — Proof of Concept Report

What's in the report

A single document covering the Fox Taylor scheduling system, with enough evidence that any engineer can verify the POC ran as described.

- Platform identification** — what software powers Fox Taylor's scheduler, its public API surface (if any), and which selectors / endpoints we depend on.
- Booking funnel map** — annotated screenshots of every page in order, with the data we read from each and the data we wrote to each.
- Service catalog mapping** — table mapping our internal triggers (recall IDs, DTCs, interval codes) to Fox Taylor's displayed service names.
- One full booking transcript** — Playwright trace + screenshots + timing for one successful appointment, with the dealer's confirmation number visible.
- Failure modes observed** — every error, popup, captcha, timeout, or unexpected redirect we hit, with how we handled it.
- Performance numbers** — wall-clock time per step, total time per booking, dealer page load times.

ACCEPTANCE CRITERIA

- One real Fox Taylor service appointment booked by the bot, confirmed by the dealer's own system, witnessed by the Fox Taylor service manager.
- Report delivered as a self-contained HTML/PDF that a non-technical reader can follow.
- All credentials, tokens, and customer data redacted.

4. Deliverable B — Repeatability Demonstration

What we demonstrate

Booking once is a stunt. Booking the same way *five times in a row* with different inputs is a system. The repeatability run proves the automation is deterministic, not lucky.

- Five bookings, back to back**, varied across:
 - Two service types (e.g., "recall remediation" + "30K maintenance")
 - Three different days/times in the dealer's calendar
 - Three different vehicles (different VINs/year/make/model)
- Per-booking record** — Playwright trace, dealer confirmation number, elapsed seconds, screenshots of the dealer's customer-facing confirmation email.
- Live re-run** — one booking executed end-to-end on a screen share with the Fox Taylor team watching.
- Cancellation log** — proof that all five test appointments were cleaned up (called and canceled by an operator), so we don't pollute the dealer's day.

ACCEPTANCE CRITERIA

- 5 of 5 bookings succeed without manual intervention.
- Median booking time under 60 seconds.
- Zero double-bookings, zero orphaned partial submissions.
- Live demo run completes successfully on first attempt.

5. Risks & mitigations

CAPTCHA OR BOT DETECTION

Some dealer schedulers run reCAPTCHA or Cloudflare Turnstile.
Mitigation: identify in reconnaissance; if blocking, route through a captcha-solver service or request Fox Taylor whitelist our IP.

SITE CHANGES BREAK SELECTORS

Dealer sites get rebranded. Mitigation: page-object pattern centralizes selectors; nightly canary run flags breakage within 24h.

TERMS-OF-SERVICE / LEGAL

Some dealer ToS prohibit automated access. Mitigation: get written consent from Fox Taylor for the POC. For production, negotiate a partnership or formal API.

REAL BOOKINGS AFFECT REAL CUSTOMERS

Mitigation: every test booking is on a Guidepoint-owned phone/email. Operator cancels each by phone within the same business day.

SLOT INVENTORY RACE CONDITIONS

The slot we read may be taken before we book. Mitigation: re-verify availability inside the submit transaction; if lost, fall back to next slot and notify Kate to renegotiate with the customer.

RATE LIMITS / IP BLOCKS

Mitigation: throttle to one session at a time during POC; coordinate with dealer IT if blocked.

6. Effort & schedule

WEEK	WORK	OUTPUT
W1 days 1–2	Reconnaissance of Fox Taylor scheduler	Platform identification + funnel map
W1 days 3–5	Read-only automation (Steps 1–3)	Open-slots fetcher (safe to run)
W2 days 1–4	Write automation (Steps 4–5) with operator review on first booking	End-to-end booking pipeline
W2 day 5	Repeatability run + report writeup	POC report + repeatability evidence

7. Coordination required from Fox Taylor

- Written consent for automated access to their public scheduler for POC purposes.
- Five test customer/vehicle records (or permission to use Guidepoint-created test customers).
- Service manager available for ~30 minutes on the day of the live demo run.
- A point of contact in their dealer IT in case our traffic looks suspicious.

8. What this enables next

- Work Package 02:** reschedule + cancel flows against Fox Taylor.
- Work Package 03:** generalize to a second dealer on the same scheduling platform — measure the marginal effort (the real test of repeatability).
- Work Package 04:** abstraction layer — one common interface, multiple dealer adapters underneath.
- Production:** queue + worker model, retries, monitoring, captcha service, observability — wraps the per-dealer automations into a reliable booking service Kate can call.